

SketchUp: Modeling for Beginners

Instructor: Menglei Zhong | 4 Class Hours

Course Outline

课程安排

Section 环节	Topic 主题	Content 具体内容
1	Preparation 准备	Initial settings + View navigation 初始设置 + 视图导航
2	CAD → Walls CAD → 墙体	Import drawings, create faces, extrude walls 导入图纸、封面、把墙拉起
3	Doors & Windows 门洞与窗户	Cut openings, create windows, copy components 开门洞、做窗户、组件化复制
4	Curved-Surface Models 曲面模型	Analyze and decompose forms; model and assemble regular curved surfaces 模型分析、拆解、规则曲面模型建模组装
5	Final Output 收尾出图	Materials, export images + plugin overview 材质、导出成果图 + 插件速览

01

Getting Started: Initial Software Setup

准备工作：软件的初始设置

Three essentials: Set units → Configure shortcuts → Navigate views

三个动作：设单位 → 设快捷键 → 转换视角

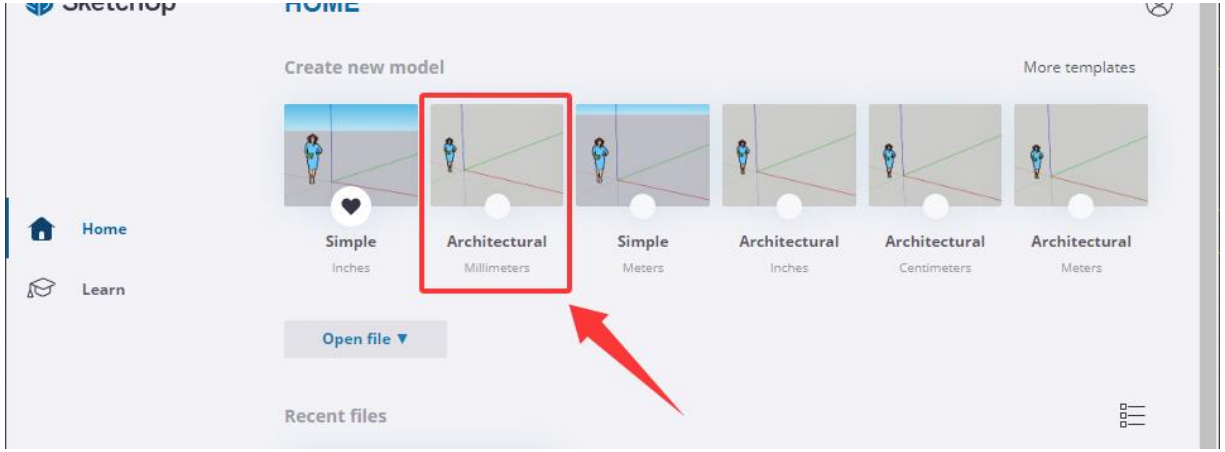
Steps in this section: 01 Setup | 02 Shortcuts | 03 View navigation

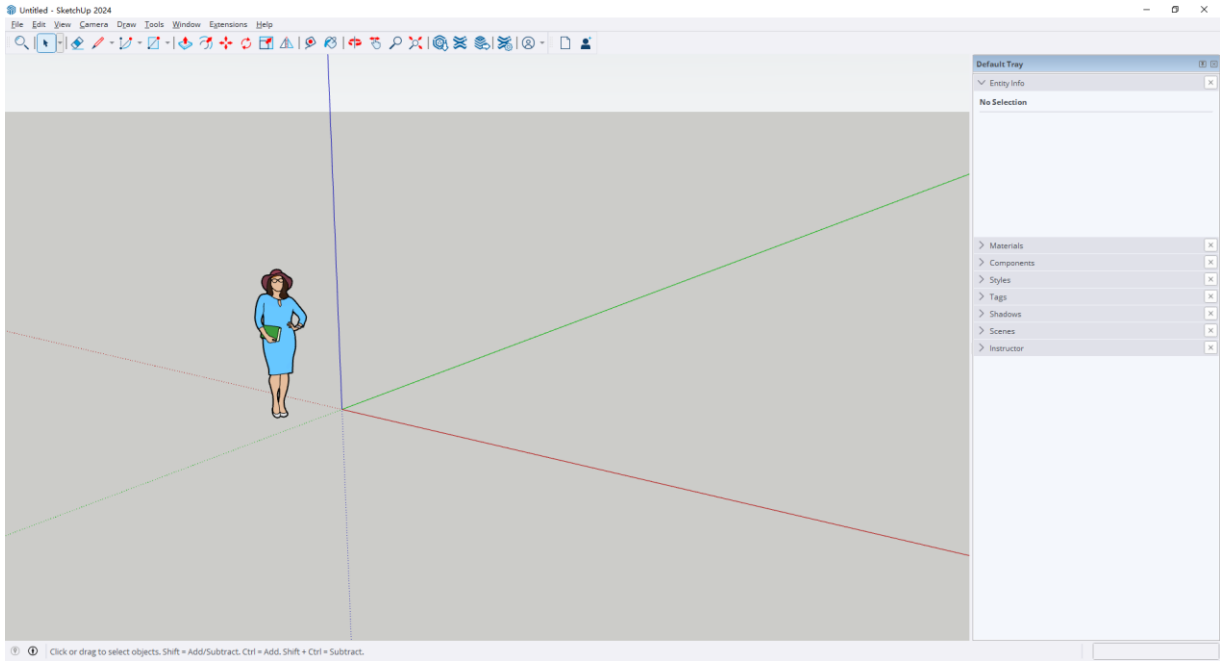
本环节步骤：01 初始设置 | 02 快捷键 | 03 视图导航

STEP 01

Initial Setup: Choose a Millimeter Template and Explore the Interface

初始设置：选毫米模板，认识界面





What to Do

你要做的

① Create a new file; choose “Architectural - Millimeters” template.
② View → Toolbars; enable “Large Tool Set”.
③ Window → Model Info → Units; confirm millimeters.

① 新建文件，模板选「建筑设计 - 毫米」
② 菜单 视图 → 工具栏，勾选「大工具集」
③ 窗口 → 模型信息 → 单位，确认是毫米

Remember the Three Axes:
Red X | Green Y | Blue Z (height)
Front/back faces: White = front; Blue = back

记住三轴：
红 X | 绿 Y | 蓝 Z（高度）
正反面：白色=正面，蓝色=反面

Common mistake: Choosing meters or inches makes every dimension wrong. Check units before modeling.

常见坑：模板选成「米」或「英寸」，后面所有尺寸都会出错——动工前先看一眼单位。

STEP 02

View Navigation
视图导航

Orbit 环绕观察
Hold the middle mouse button and drag to rotate around the model.

按住鼠标中键拖动，围着模型转圈看

Pan 平移
Shift + middle mouse drag to move your view.

Shift + 鼠标中键拖动，把画面挪到要看的位置

Zoom 缩放
Scroll to zoom; Z to zoom in; Shift+Z for Zoom Extents.

滚动滚轮推近拉远；Z 框选放大，Shift+Z 看全图

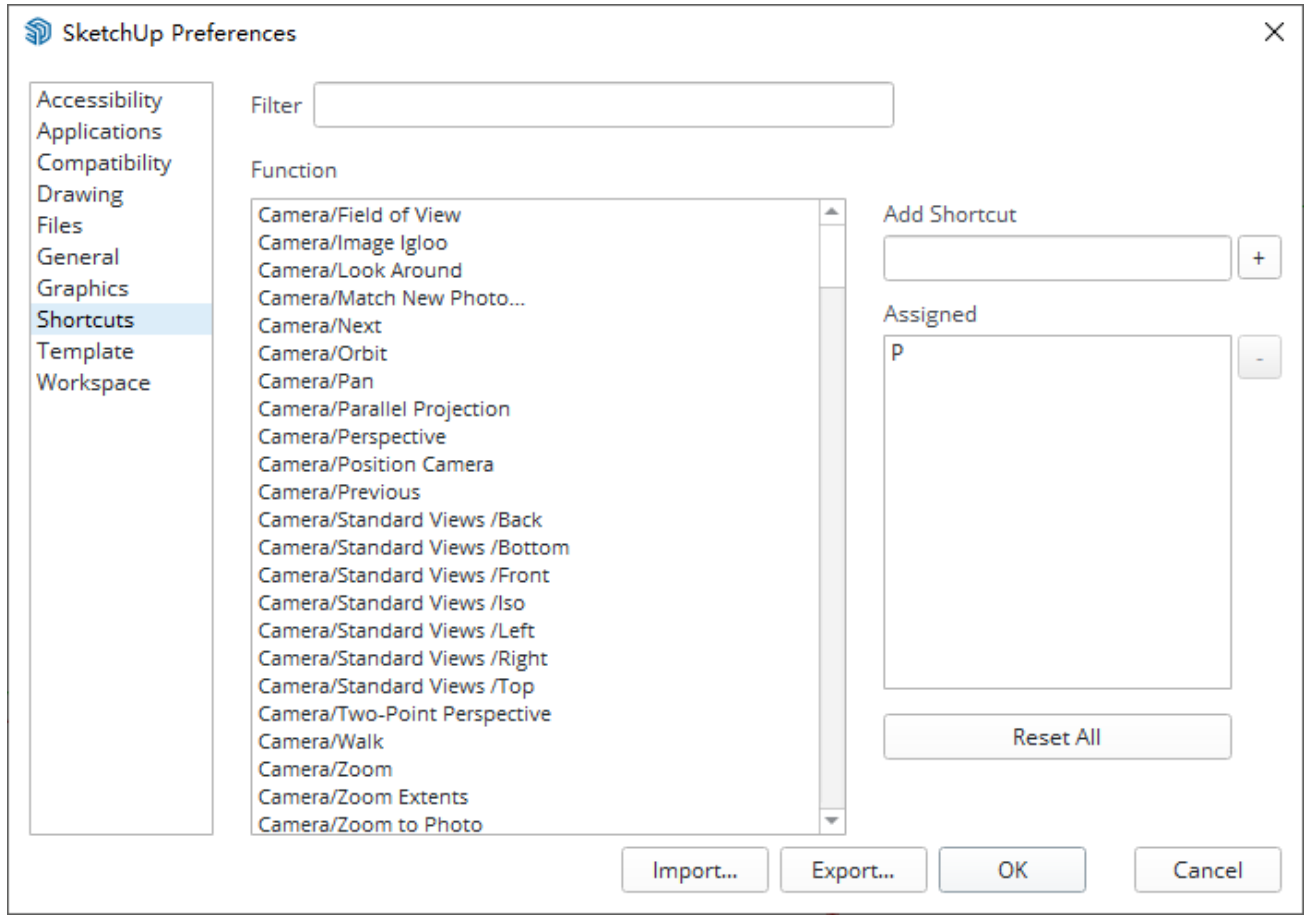
Practice: Draw a rectangle → Push/Pull (P) into a box → Try orbiting, panning, and zooming three times each.
跟做练习：画一个矩形 → P 拉成盒子 → 用中键转着看、Shift+中键平移、滚轮缩放，各试 3 次

Standard Views 标准视图	Camera → Standard Views: Top / Front / Left; use for plan alignment 相机 → 标准视图：顶 / 前 / 左，齐图纸时用	Perspective / Parallel 透视 / 平行投影	Perspective for renders; parallel for diagrams 出效果图用透视，分析图用平行投影
Find the Model 找回模型	Shift+Z: Zoom Extents Shift+Z 一键让模型充满视图		

STEP 03

Keyboard Shortcuts

快捷键设置



Essential Shortcuts

最小快捷键集

Space 空格	Select tool (most used) 选择工具（最常用）
L	Line 直线
R	Rectangle 矩形
P	Push/Pull 推拉
M	Move (+Ctrl to copy) 移动（+Ctrl 复制）
F	Offset 偏移
E	Eraser 橡皮擦
G	Make Component 创建组件

Custom setup: Window → Preferences → Shortcuts

自定义路径：窗口 → 系统属性 → 快捷键

02

From CAD to Walls: Bringing a Floor Plan into 3D

从 CAD 到墙体：图纸站起来

2D to 3D: Clean CAD → Import → Group & Lock → Create Faces → Extrude Walls

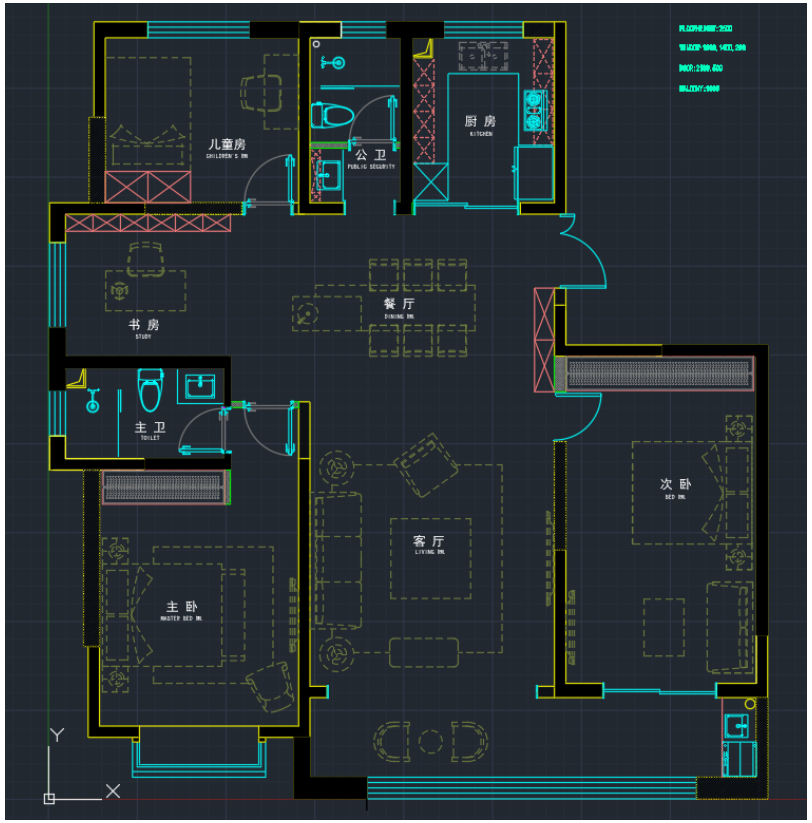
从二维到三维：清理 CAD → 导入 → 成组锁定 → 封面 → 推拉成墙

STEP 04

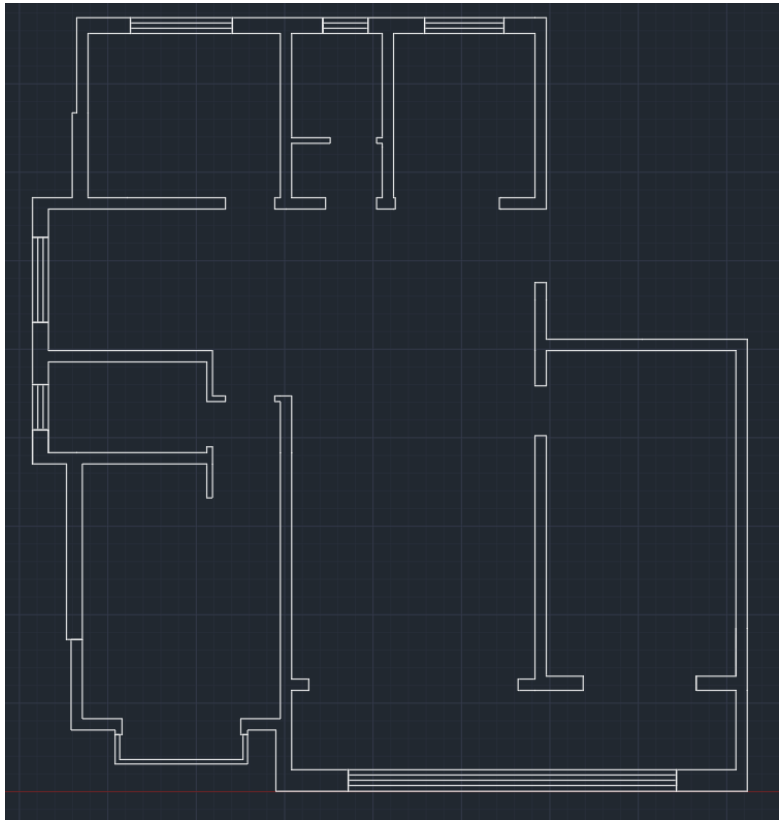
Clean the CAD: Keep Only Wall and Door/Window Opening Lines

清理 CAD：只留墙线和门窗洞口线

Before cleanup (original CAD)
清理前（CAD原图）



After cleanup (wall and opening lines only)
清理前（只留墙体门窗线）



Clean Up the Drawing
清理图纸

① Delete dimensions, text, hatches, and furniture.
② Select all → FLATTEN.
③ Check that all lines connect correctly.
④ Use JOIN to check connected lines.
⑤ Remove unused layers and save as a new file.

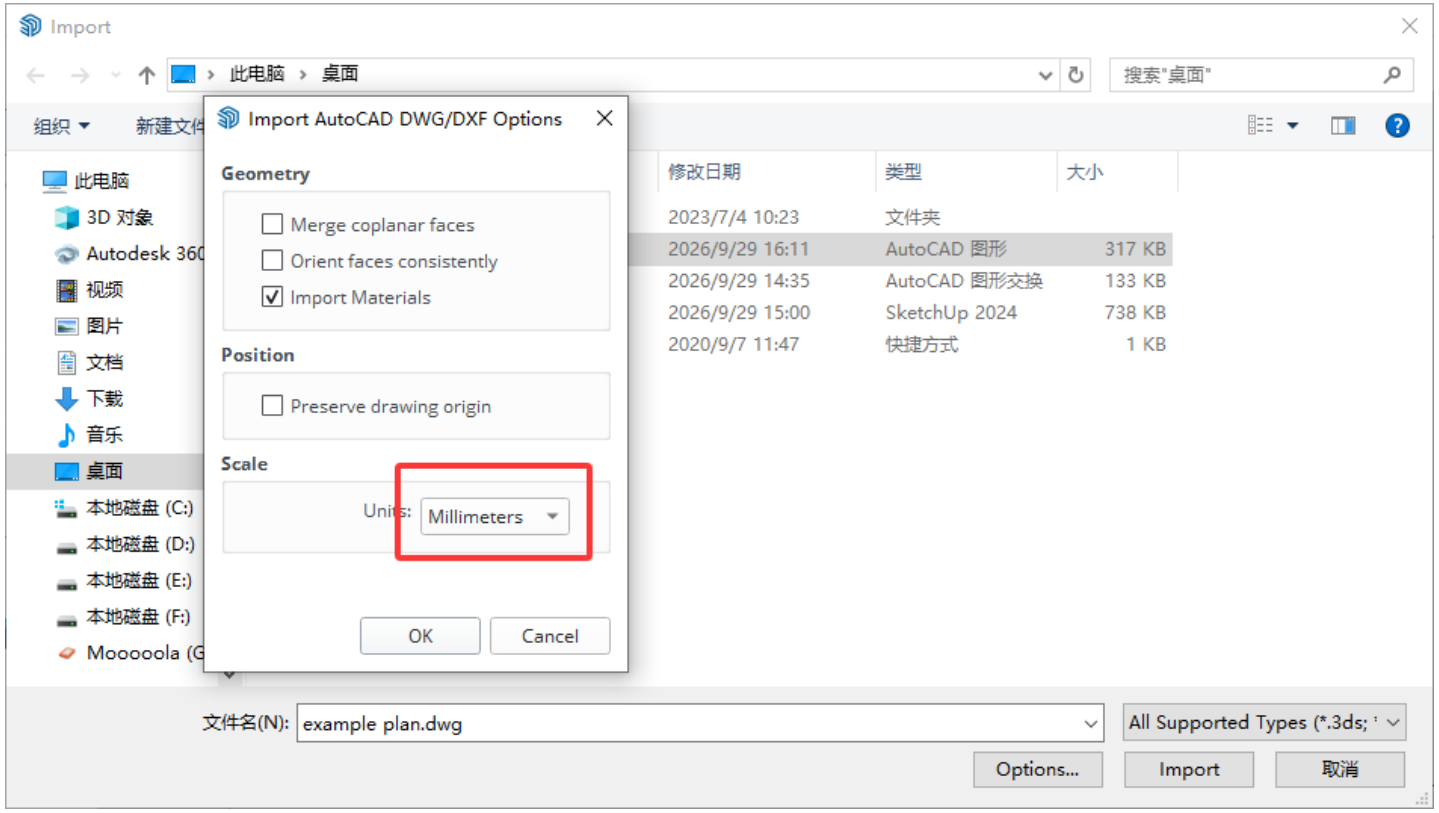
① 删掉标注、文字、填充、家具
② 全选 → flatten
③ 确认所有的线是否都连接正常
④ 可以通过join来检查线的连接
⑤ 清理无用图层，另存新文件

Why flatten?
Lines on different planes cannot form faces after importing into SketchUp.
为什么归零：
线条不共面，导入 SU 后永远封不上面

STEP 05

Import into SketchUp: Set Units to Millimeters in Options

导入 SU：先点「选项」，单位选毫米



Procedure

操作路径

① File → Import; select DWG as the file type.
② Click "Options": Units = Millimeters;
③ Select the drawing → Import.

① 文件 → 导入，文件类型选 DWG
② 点右下角「选项」：单位 = 毫米
③ 选中图纸文件 → 导入

Common mistake: Skipping Options may interpret inches as millimeters, enlarging the model by 25.4×.

最常见的错：跳过「选项」直接导入，图纸被按「英寸」解读，整个模型放大 25.4 倍

STEP 06

Group and Lock the CAD Underlay

底图成组 + 锁定

① Select all lines ① 全选线稿	② Make a group ② 创建群组	③ Right-click to lock ③ 右键锁定
Select with a box / Ctrl+A 框选 / Ctrl+A 选中全部导入线条	Right-click → Make Group (Ctrl+G); blue bounding box appears 右键 →「创建群组」（Ctrl+G），出现蓝色包围盒	Right-click → Lock; avoid accidentally moving CAD lines 右键 →「锁定」，之后描图时再也不会误选、误移底图

Why Group Before Modeling?

为什么要先成组再建模

Geometry touching in SketchUp automatically merges. Without grouping the CAD underlay, new wall lines will stick to its edges and make later edits difficult.

Golden rule: Group separate elements before refining them.

SU 里所有「粘」在一起的几何体会自动合并。底图不成组，你画的第一根墙线就会和 CAD 线稿焊死，后面想改墙，连图纸一起被扯坏。
建模铁律：不同构件，先成组，再细化。

Group or Component?

群组 OR 组件

Group 群组	Component 组件
Protect separate objects 独立对象打包保护	Linked editing of repeated objects 重复元素关联修改
Editing one does not affect others 改一个，其他不受影响	Editing one updates all instances 改一个，副本全部联动
CAD underlay, walls, roof 底图、墙体、屋顶	Windows, railings, furniture 窗户、栏杆、桌椅

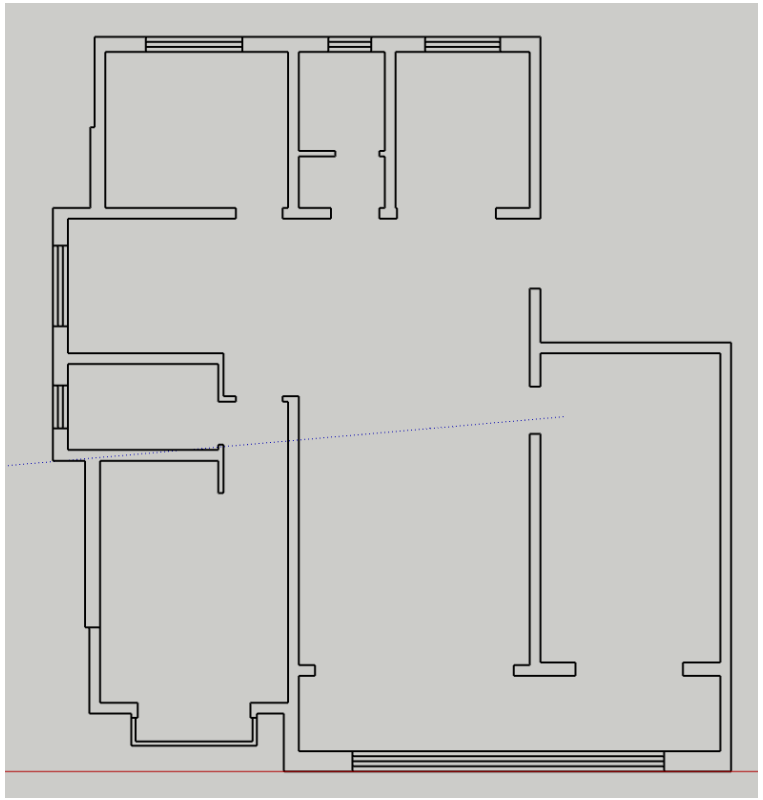
Shortcut: Ctrl+G creates a group. To edit the underlay again: right-click → Unlock.

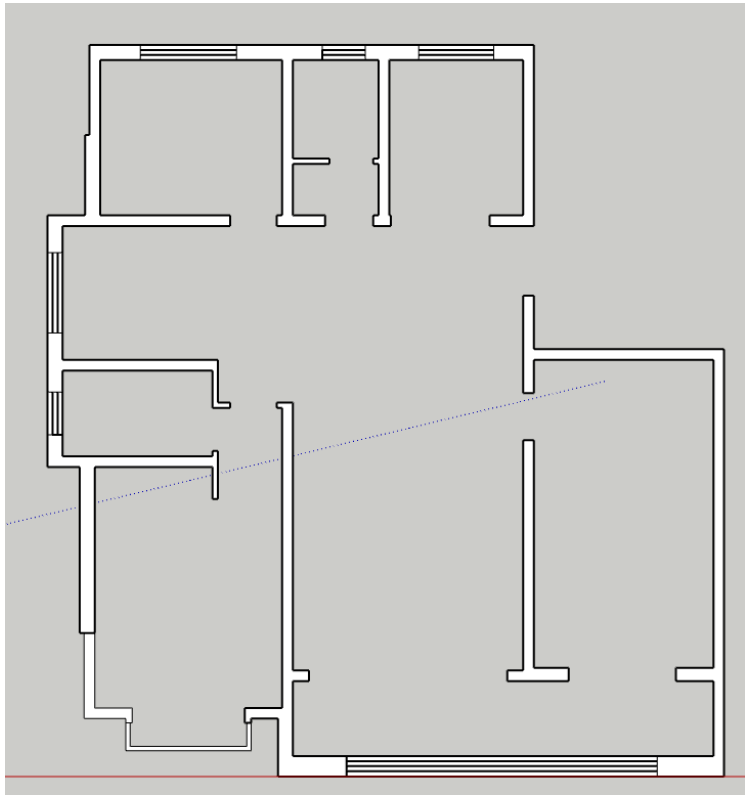
快捷键：Ctrl+G 创建群组解锁：右键 →「解锁」可恢复编辑底图

STEP 07

Create Faces: Trace Closed Edges over the CAD Underlay

封面：沿底图描线，闭合即成面





What to Do

你要做的

① Press L for the Line tool.

② Start at a wall intersection and trace each edge.

③ Close the loop at the starting point to create a white face.

① 按 L 使用直线工具
② 从墙线交点开始，逐边描线
③ 回到起点闭合 → 出现白色的面

Why a Face May Not Form:

- Lines do not form a closed loop.
- Line ends extend past intersections.
- Lines are not coplanar (CAD not flattened).

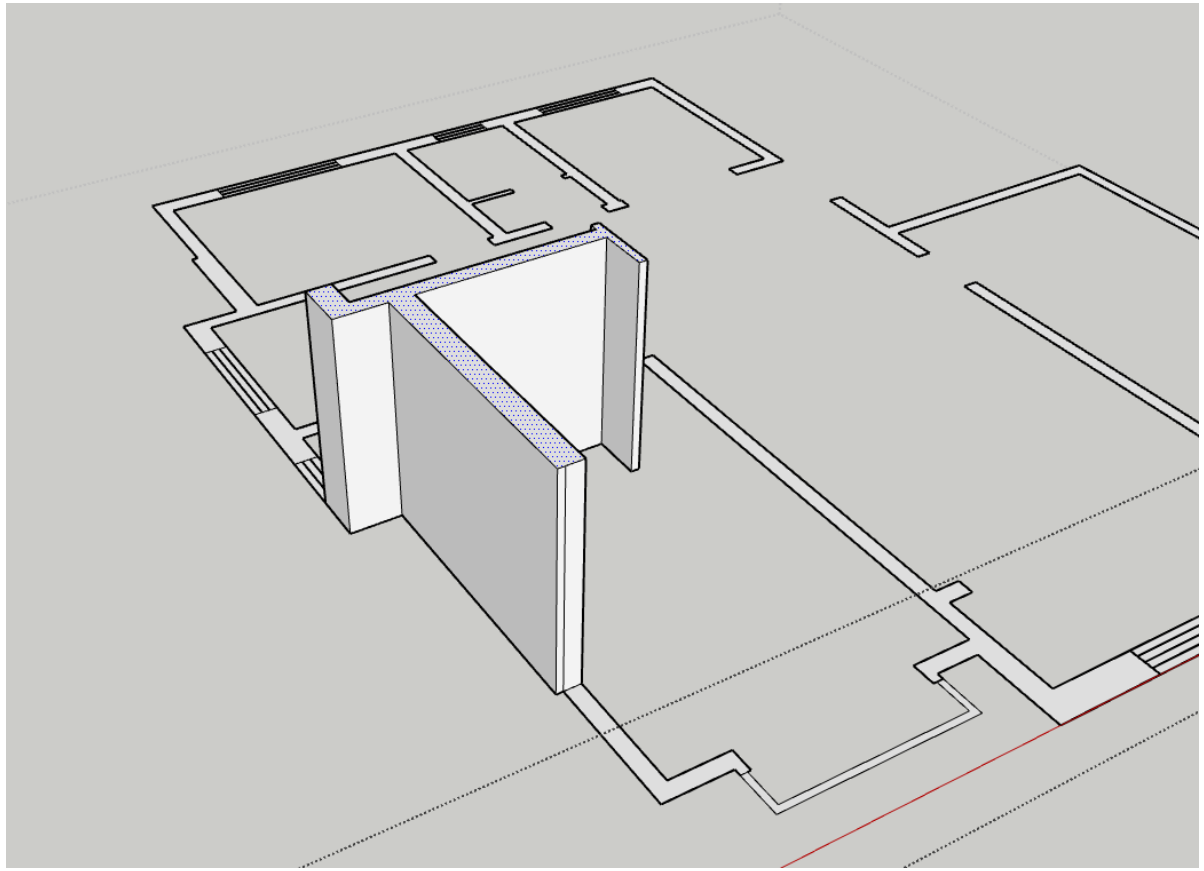
Fix: Zoom in (Z), fill gaps, or erase extra edges.

封不上面的三种原因：
· 线条没闭合（有缺口）
· 线头超出交点
· 线条不共面（CAD 没归零）
对策：Z 放大找缺口，补线或删多余线头

STEP 08

Extrude Walls: One Push/Pull (P) Brings Walls into 3D

墙体拉伸：P 键一拉，墙就站起来了



What to Do
你要做的

① Press P and click the wall face.
② Drag upward, type 3000, and press Enter (no need to click the input box).
③ Double-click other faces to repeat the last height.

① 按 P，点击封好的墙面
② 向上拖动，键盘输入 3000，回车（不用点输入框）
③ 其他墙面：直接双击 = 重复上次高度

Push/Pull Essentials:
- Click face → Drag → Enter value.
- Double-click = repeat the last distance.
- Ctrl = copy the pushed/pulled face.

Then select all walls → Ctrl+G to group.

推拉三字诀：
· 点面 → 拖动 → 敲数字
· 双击 = 重复上次
· 按 Ctrl = 复制推拉（加厚体块时用）
完成后：框选全部墙体 → Ctrl+G 成组

Check face orientation: Inside walls should appear white (front). For blue back faces, right-click → Reverse Faces. Some renderers may omit reversed faces.
正反面检查：转到房子里面看——墙面应为白色（正面）。出现蓝色反面：选中 → 右键 →「反转平面」。反面朝外会导致部分渲染器丢面。

IN-CLASS PRACTICE

Task Checklist (Complete in Order)
任务清单（按顺序做）

✓	Action 动作	Completion Check 完成标准
1	Import “Residential Plan - Cleaned.dwg” (units: mm) 导入「住宅平面-已清理.dwg」（选项→毫米）	Measure wall; correct dimensions 卷尺量墙，尺寸正确
2	Select lines → Ctrl+G → Lock 线稿全选 → Ctrl+G 成组 → 锁定	Blue bounding box appears 出现蓝色包围盒
3	Trace with L to form outer wall faces L 描线，给外轮廓封面	White faces appear 出现白色面
4	Push/Pull (P) to 2600 mm; double-click others (balcony: 1000) P 推拉至 2600 mm，其余面双击推拉（阳台1000）	Wall heights are consistent 整圈墙高一致
5	Group walls; check front/back faces 墙体成组，检查正反面	All interior faces are white 内侧看全为白色
6	Ctrl+S: save as Name - Residence.skp Ctrl+S 保存：姓名-住宅.skp	File saved 文件已保存

Stuck? Check These Three Things First
卡住了？先自查这三处

- Model is much too large/small → Re-import using millimeters.
- Face will not form → Zoom in for gaps/extra edges, or redraw that edge.
- Push/Pull does not work → Select the face, not the locked CAD underlay.

· 尺寸差很多倍 → 导入时没设毫米，重新导入
· 封不上面 → Z 放大找缺口/线头，或重画该边
· 推拉没反应 → 面没选中，或点在底图组上了（底图已锁定，点不动是正常现象）

03

Door and Window Openings

门洞与窗户

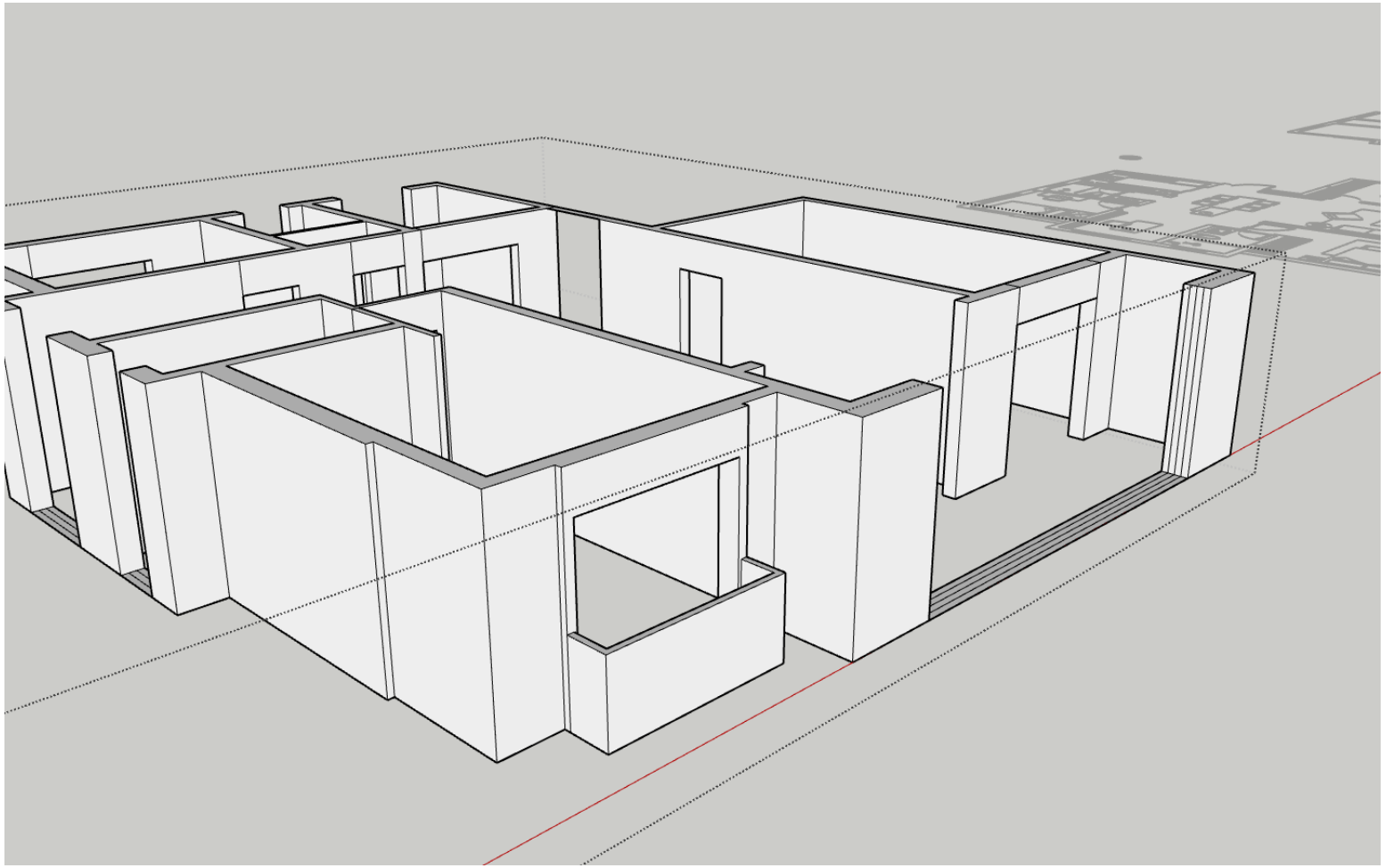
Door = Rectangle + Push/Pull Through; Window = Position + Offset + Component

门洞 = 画矩形 + 推拉贯穿; 窗户 = 定位 + 偏移 + 组件

STEP 09

Door and Window Openings: Draw Rectangles and Push/Pull

门洞+窗洞：画矩形，推拉出墙体



What to Do

你要做的

Door Opening

门洞

① Door height: 2100.

② Draw a rectangle at the top; Push/Pull downward by 500.

③ Press E to erase remaining edges in the opening.

① 门高度：2100
② 画出顶部矩形，向下推拉500
③ E 删掉洞口里的残留线

Window Opening

窗洞

① Sill height: 900.

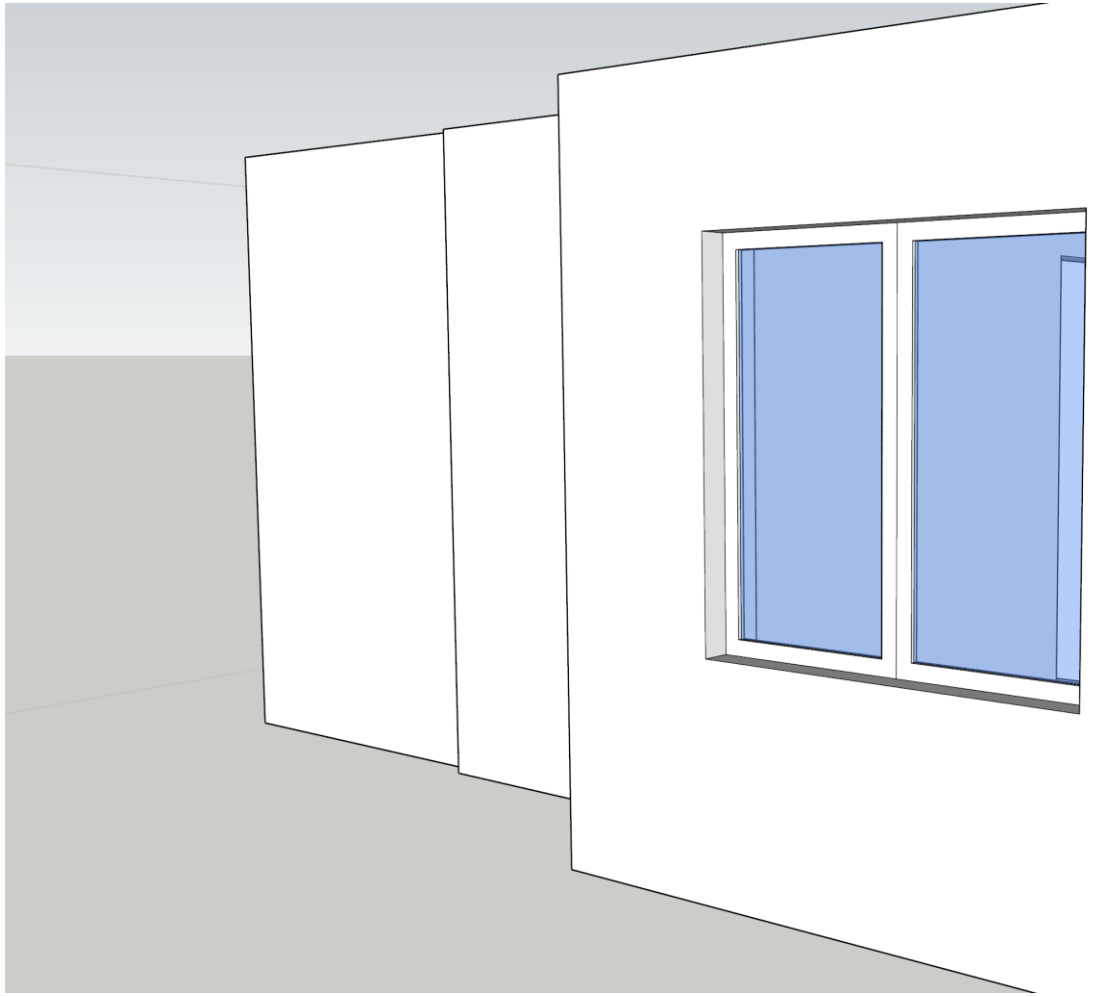
② Window height: 1500.

① 窗台高度：900
② 窗高度：1500

STEP 10

Make Windows: Position First, Then Offset the Frame

窗户制作：先定位，再偏移出窗框



What to Do

你要做的

① Draw the outline of the window frame.
② Offset inward by 60 to make the frame.
③ Draw the glass inside the frame.
④ Make standard windows into named components.

① 绘制窗框轮廓线
② 向内偏移60作为窗框
③ 绘制窗框内玻璃
④ 标准化窗户可以创建组件命名

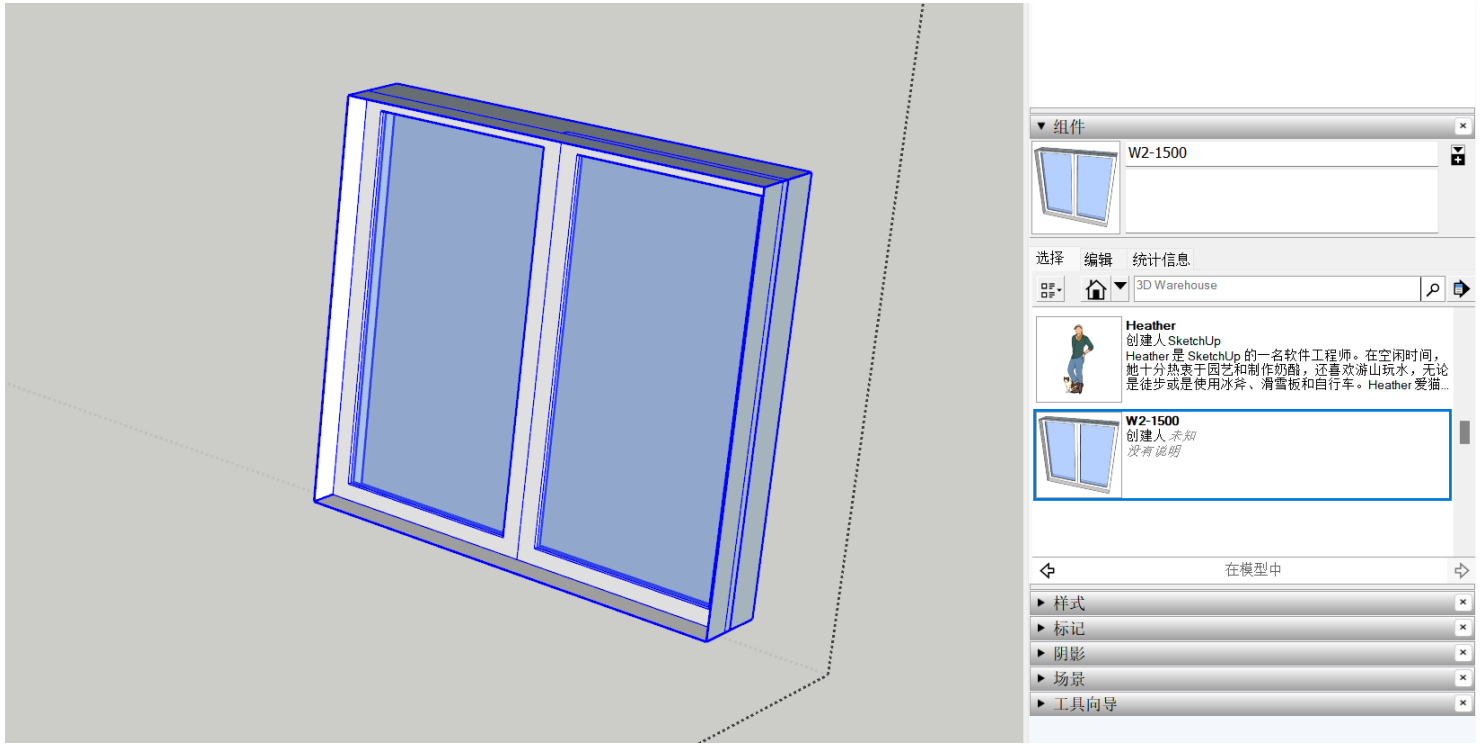
STEP 11

Components and Copying: One Window for the Whole Building

组件与复制：做一扇窗，用满整栋房

① Select the Window ① 选中整扇窗	② Make Component (G) ② 创建组件 G	③ Copy with M+Ctrl ③ M+Ctrl 复制
Edit the wall group; select frame + glass 双击进入墙组，框选窗框+玻璃	Name it “W1-600”; align axes with wall plane 命名「W1-600」，注意组件坐标轴贴在墙面上	Move copies to other windows 拖到其余窗位

Group 群组	Component 组件
Independent instances 独立打包，副本互不影响	Linked master and copies 母版关联，副本联动
Use: walls, roofs, CAD underlay 用在：墙体、屋顶、底图	Use: doors, windows, railings, furniture 用在：门窗、栏杆、家具



IN-CLASS PRACTICE

Task Checklist (Complete in Order)
任务清单（按顺序做）

✓	Action 动作	Completion Check 完成标准
1	Create door and window openings based on CAD 按底图位置绘制门洞和窗洞	Door opening goes through wall 门洞两面通透
2	Make the wall a group 墙体建立群组	Wall remains complete 墙体完整
3	Draw frame edges; Offset (F) inward by 60 绘制窗框边缘，F 偏移 60 做窗框	Frame and glass have depth 窗框玻璃有进退
4	Select window → Make Component (G) + number 整窗选中 → G 组件「编号」	All windows completed 完成全部窗户
5	Create door components similarly 相应办法制作门组件	All doors completed 完成全部门

04

Curved-Surface Models: Analysis and Decomposition

曲面模型：模型分析+模型拆解

Three-Step Form Analysis + Follow Me + Intersect Faces

曲面三步分析法 + 路径跟随 + 模型交错

STEP 12

Analyzing and Breaking Down Forms

模型拆解分析

1 Study the Form 看形体

Look at the reference: Is the shape assembled from regular curved surfaces? How can it be broken down?

对着参考图问自己：这个曲面是否是由一些规律曲面拼合而成的呢？如何进行形体的拆解？

2 Identify Control Lines 控制线

Find the key section, path, and outline curves that define the form.

找出决定造型的关键线条：截面线、路径线、轮廓线。

3 Plan the Sequence 排顺序

Decide which lines to draw first, which surfaces to make next, and where to cut last.

规划先做哪条线、后生成哪个面、最后切哪里。

Follow Me 路径跟随	Intersect Faces 模型交错	Rotate Array 旋转阵列	Sandbox Tools 沙盒工具
Sweep a profile along a path (pergolas, vaults, handrails) 截面沿路扫出曲面（廊架、拱顶、扶手）	Intersect solids for openings in curved surfaces 两体相交切出洞口（曲面开窗、开洞）	Copy around an axis (petals, domes) 绕轴复制出环形构成（花瓣、穹顶构件）	Shape terrain from grids (landforms, projections) 网格起伏做地形（微地形、曲面投射）

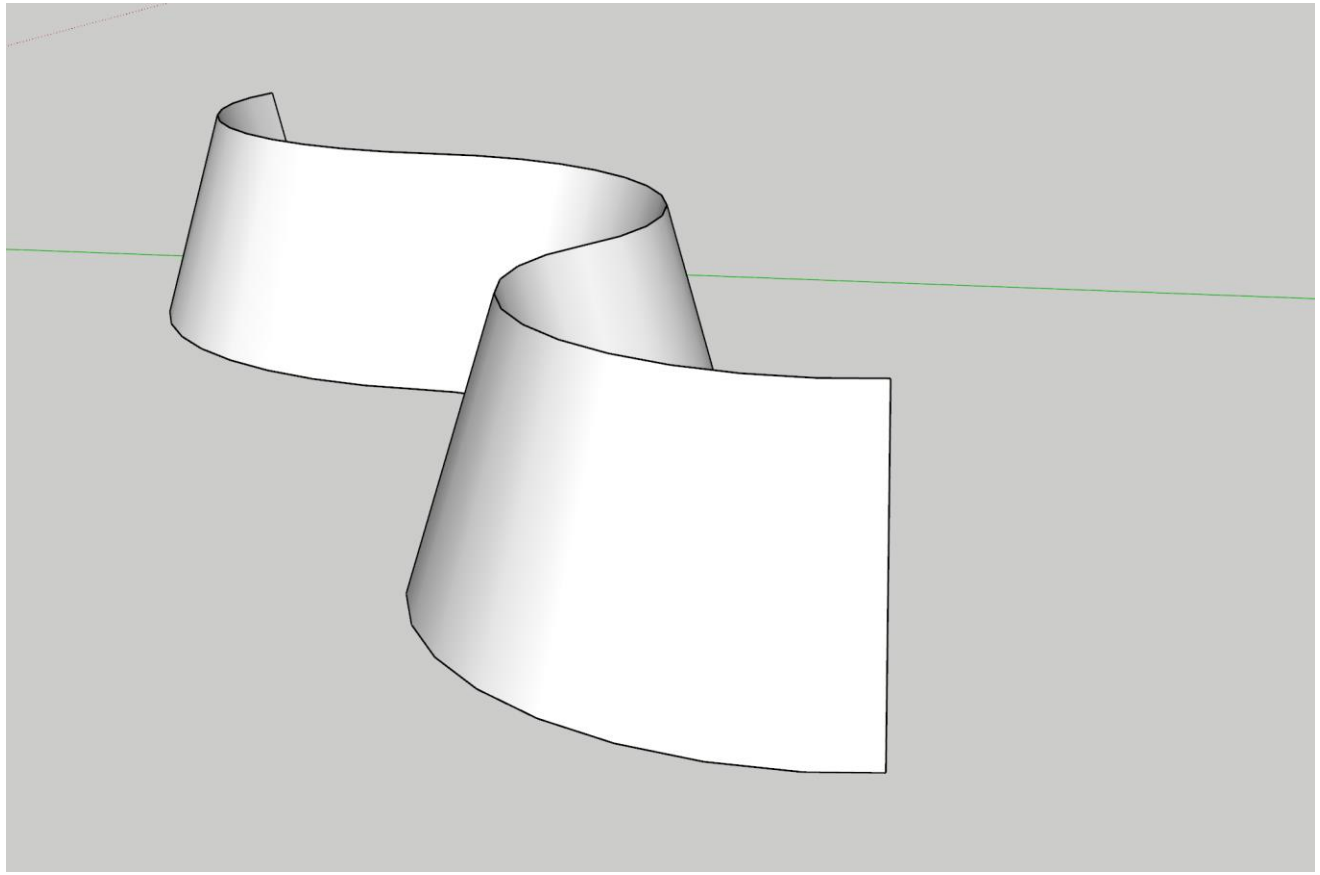
Method reference: Organic modeling without plugins https://www.bilibili.com/video/BV1ES4y1R78X/?spm_id_from=333.337.search-card.all.click&vd_source=c6532f2c9ea9848fd1afcc0bb37fdee1

方法论来源：无插件异形建模思路 https://www.bilibili.com/video/BV1ES4y1R78X/?spm_id_from=333.337.search-card.all.click&vd_source=c6532f2c9ea9848fd1afcc0bb37fdee1

STEP 13

Follow Me: Sweep a Profile Along a Path to Make a Surface

路径跟随：截面沿着路径「扫」出曲面



Follow Me

路径跟随

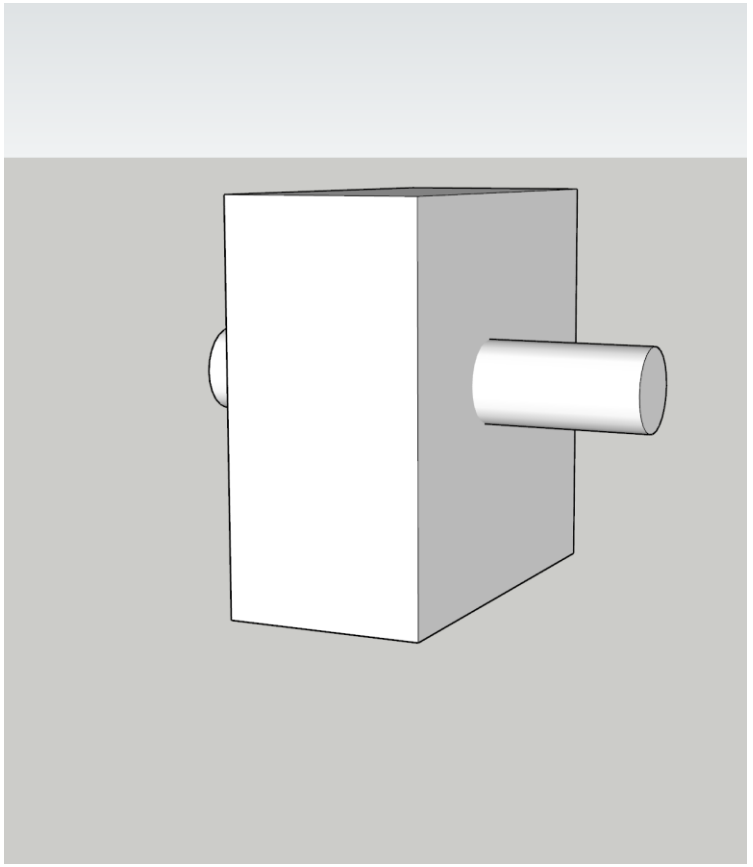
① Draw the profile and path.
② Activate the Follow Me tool.
③ Click the profile and follow the path to its endpoint.

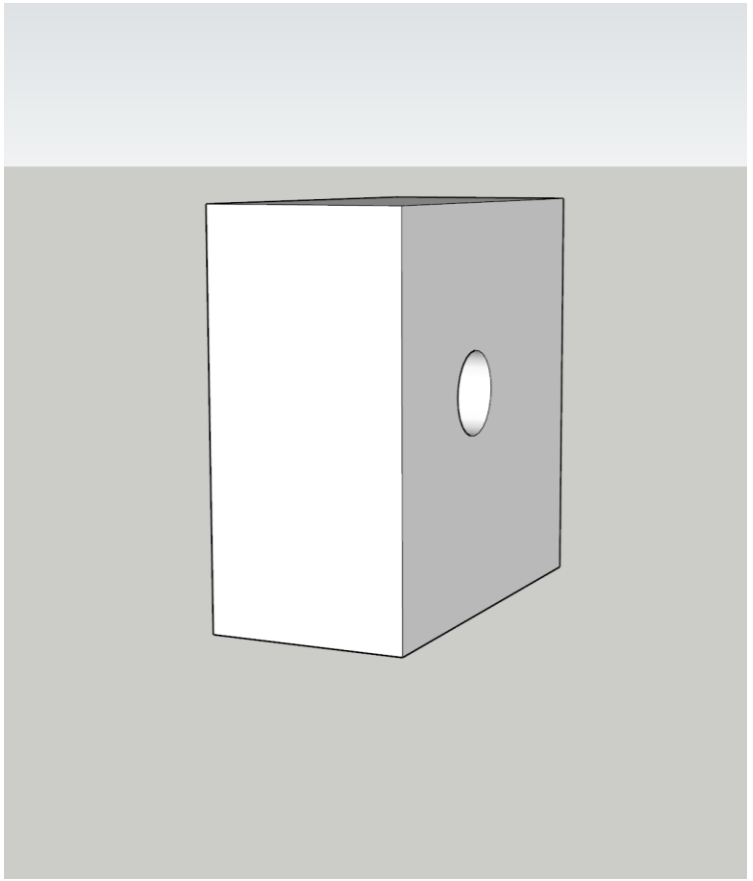
① 绘制形体的截面与路径
② 使用路径跟随工具
③ 点击截面 - 沿着路径选择终点

STEP 14

Intersect Faces: Boolean-style Openings in Curved Surfaces

模型交错：SU 版「布尔运算」，曲面开洞





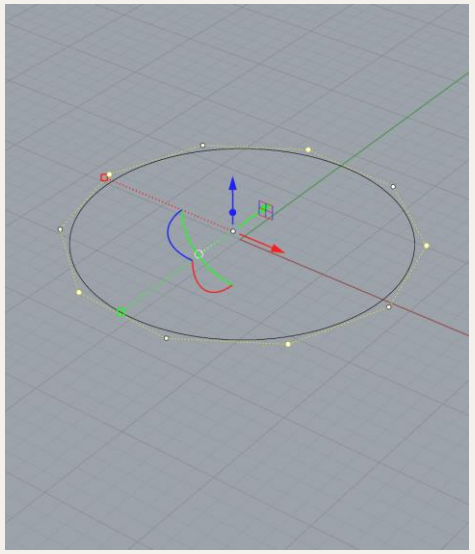
Procedure
操作顺序

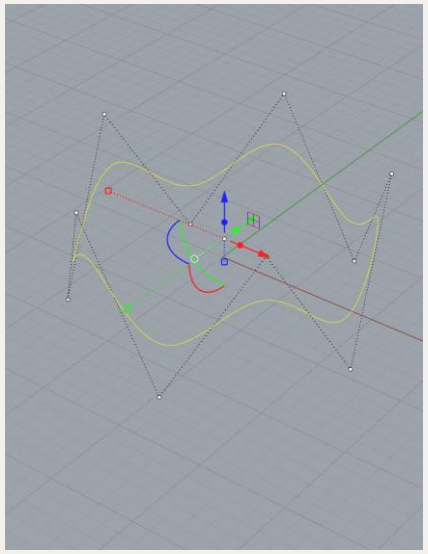
① Intersect two solids (e.g., a cylinder passing through a curved wall).
② Tools → Solid Tools.
③ Create intersection edges; use E to erase unwanted faces and reveal the opening.

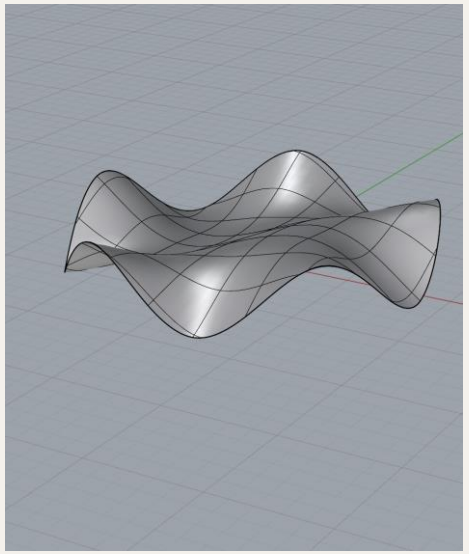
① 让两个体块相交（如圆柱贯穿曲面墙）
② 工具 → 实体工具
③ 交线自动生成，E 删掉不要的面 → 洞口出现

Rhino Surface Modeling
Rhino曲面建模

Rhino is recommended for freeform and irregular geometry.
自由异形建模建议使用Rhino软件









Create Freeform Surfaces in Rhino
① Draw basic outline curves.
② Rebuild curves and adjust control points.
③ Use surface creation tools.

Rhino 生成自由曲面
① 绘制基础轮廓线
② 重建曲线，调整控制点
③ 使用曲面生成工具

05

Finalizing and Exporting

收尾出图

Materials + Shadow Scenes + JPG Export + Plugin Overview

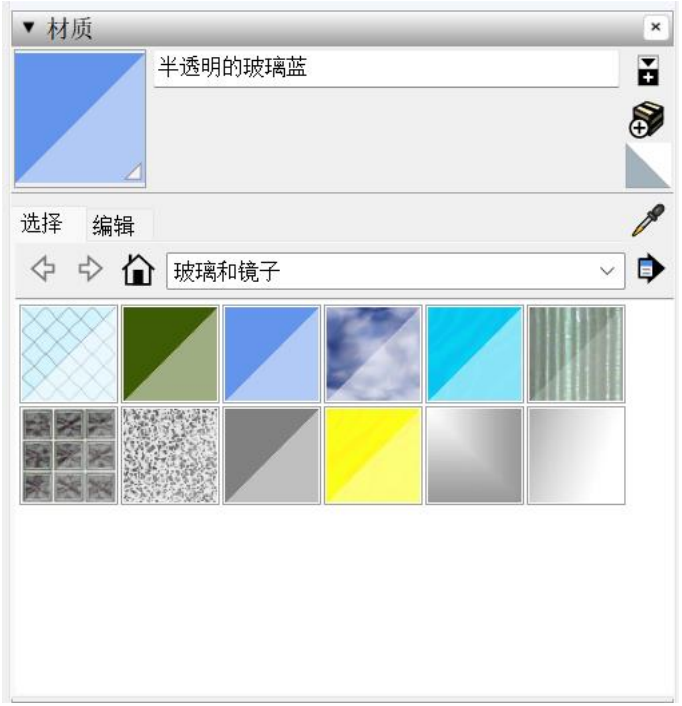
材质 + 阴影场景 + 导出 JPG+插件速览

STEP 15

Applying Materials and Textures

材质贴图





Assign Materials by Building Element

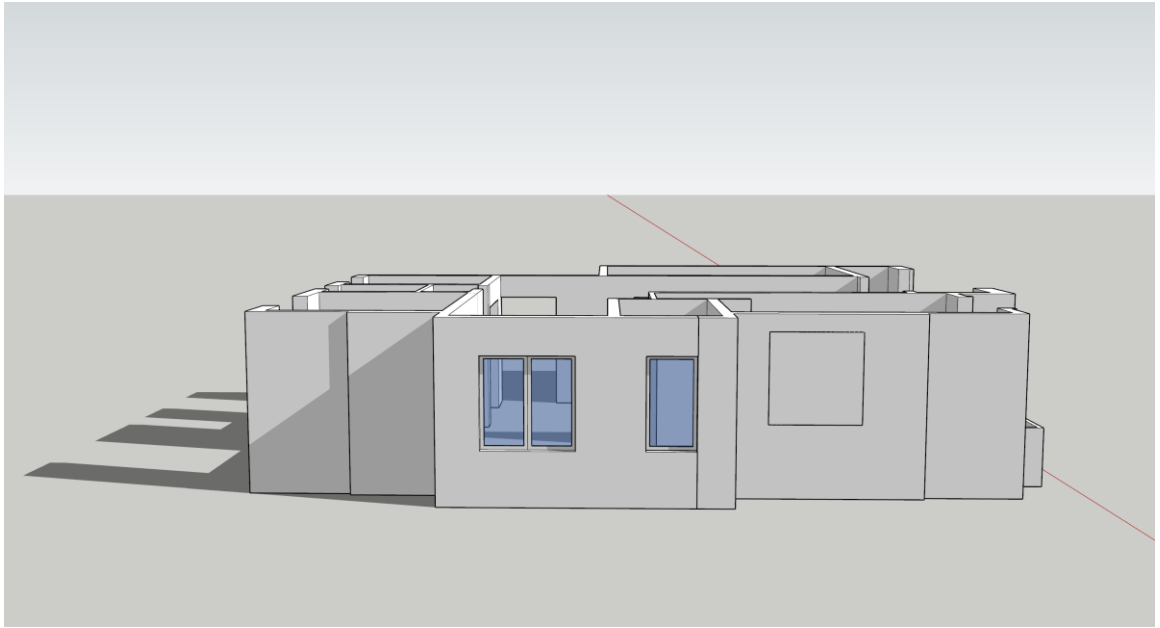
材质赋予（按构件赋材质）

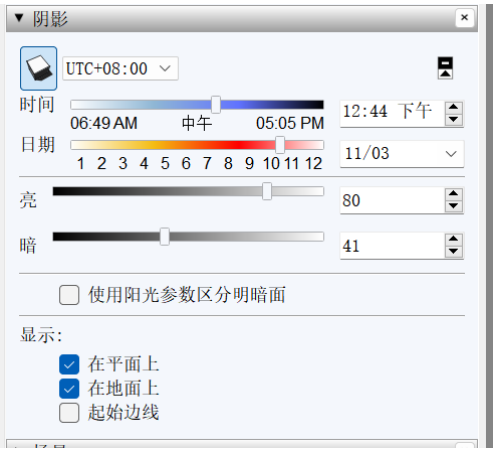
Roof 屋顶	Roof tile / metal panel 瓦片 / 金属板材质
Exterior Wall 外墙	Paint or facing brick 涂料或面砖材质
Window Frame 窗框	Dark metal / wood finish 深色金属 / 木色
Glass 玻璃	Translucent glass (adjust opacity) 半透明玻璃（调不透明度）
Ground 地面	Grass / paving materials 草地 / 铺地材质

STEP 16

Shadows and Scenes

阴影与场景





Turn on Shadows

打开阴影

① View → Shadows for more depth in the model.
② Window → Default Tray → Shadows; adjust time/date to move the sunlight.
③ Angled light around 9–10 a.m. often makes building forms clearer.

① 视图 → 阴影，模型立刻有了体积感
② 窗口 → 默认面板 → 阴影：拖动「时间/日期」滑块，光影随之转动
③ 建筑表现常用上午 9–10 点的斜射光，明暗关系最清楚

Save Scenes (Camera Views)

保存场景（机位）

① Use middle mouse and scroll wheel to find a good angle.
② Window → Default Tray → Scenes; click “+” and name it “Eye Level – Southeast”.
③ Save two more: Bird’s-eye and Front Elevation; click scene tabs to switch views.

① 用中键 + 滚轮把模型摆到最好看的角度
② 窗口 → 默认面板 → 场景，点「+」，命名「人视-东南」
③ 再存 2 个场景：鸟瞰、正立面——之后点场景标签即可秒切机位

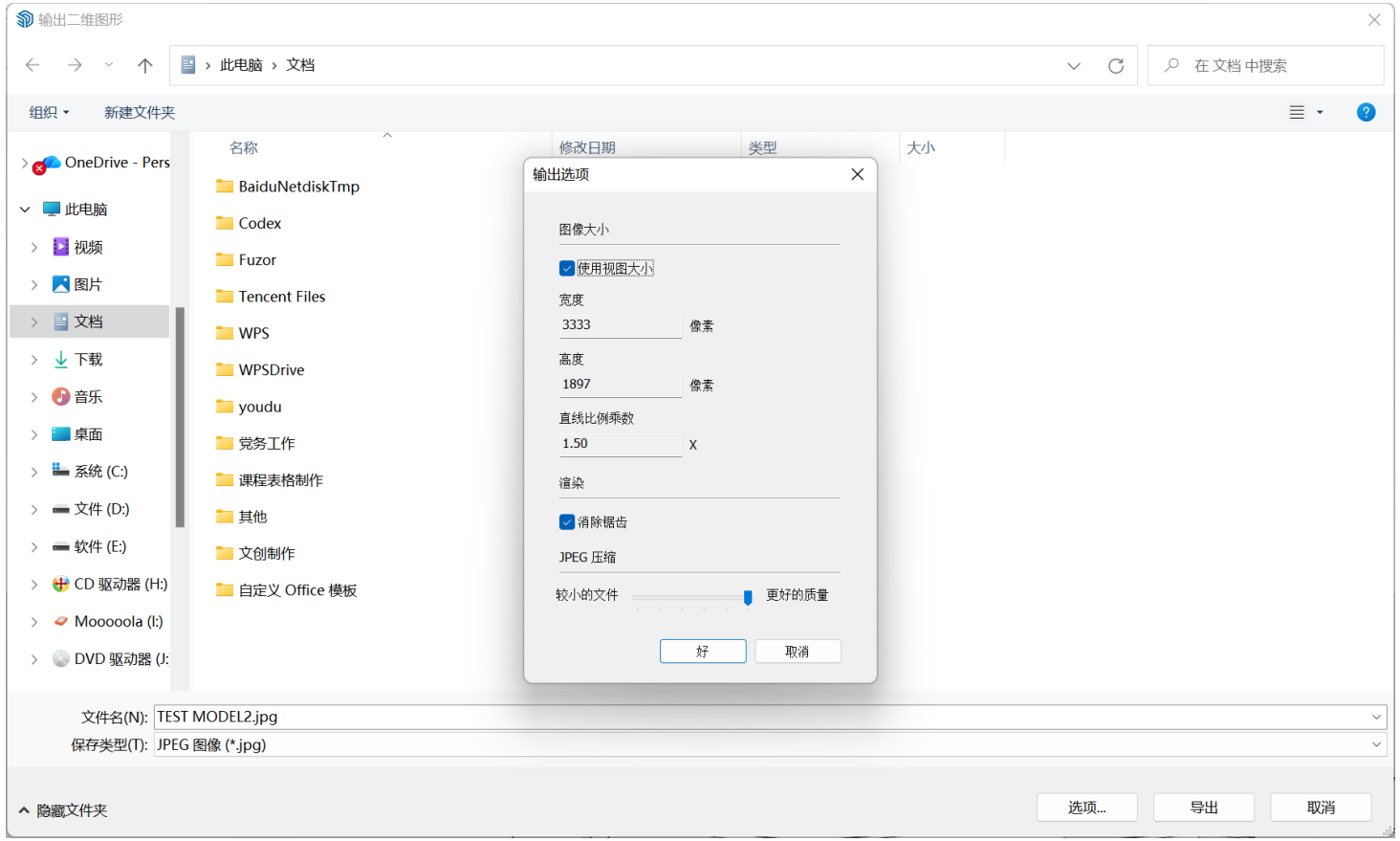
Recommended views: ① Eye Level (main image) ② Bird’s-eye (overall layout) ③ Front Elevation (proportions)

建议机位：① 人视角（主图）② 鸟瞰（看总关系）③ 正立面（看比例）

STEP 17

Export Final Images

导出成果图



Procedure

操作路径

① Click a scene tab to choose the camera view.

② File → Export → 2D Graphic; select JPG or PNG.

③ Options: Width = 3000 px; enable antialiasing.

④ Export one image per scene: Eye Level / Bird’s-eye / Elevation.

① 点场景标签，切到要导出的机位
② 文件 → 导出 → 二维图像，格式选 JPG / PNG
③ 点「选项」：宽度 3000px，勾消除锯齿
④ 3 个场景各导一张，命名：人视 / 鸟瞰 / 立面

STEP 18

Plugin Overview

插件速览

① Extension Warehouse

② SketchUcation Plugin Store

③ 坯子插件库

④ SUAPP



一键安装：坯子库官网下载管理器 → 登录 → 点「安装」即装即用（示意界面）

SketchUp plugin library interface (teaching illustration)

坯子插件库界面（教学示意图）

Category 类别	Plugin • Main Function 代表插件 · 一句话功能
Platform 平台	Piziku Library: free plugin installation platform 坯子库：免费插件平台，一键安装
Create Faces 封面	Quick Face: create faces from CAD lines 快速封面：CAD 线稿一键成面
Elements 构件	Parametric windows / stairs / railings / roofs 参数开窗 / 创建楼梯 / 栏杆生成 / 参数屋顶
Push/Pull 推拉	Joint Push Pull: thicken and extrude curved surfaces 超级推拉：曲面加厚、法线推拉
Curves 曲线	Bezier Curves: draw smooth spatial curves 贝兹曲线：画光滑空间曲线
Surfaces 曲面	Curviloft / Soap Skin: loft surface from curves 曲线放样 / 起泡：截面放样成曲面
Cleanup 清理	Cleanup: reduce model size and complexity 清理场景：模型瘦身，越用越顺

Today’s Assignment

今日课程练习

Complete a 3D SketchUp model of your own design.

✓ Clean CAD plan → Import → Create Faces → Model walls, doors, and windows.

✓ Manage repeating objects with components.

✓ Add roofs, materials, and shadow scenes; export 3000 px images.

完成个人设计的实体模型的SU三维建模

✓ 一张干净的 CAD 图纸 → 导入 → 封面 → 墙体门窗的完整流程

✓ 用组件管理重复构件

✓ 屋顶、材质、阴影场景，导出 3000px 成果图